

Breast Cancer Classification Using Machine Learning

Exploratory Data Analysis, Feature Engineering,
and Model Evaluation

Prepared by

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Project Focus

- Medical data analysis and binary classification
- Data preprocessing, exploratory data analysis, and feature engineering
- Machine learning model training and performance evaluation
- Comparison of classification models across datasets

Technologies

Python • Pandas • NumPy • Scikit-learn • TensorFlow/Keras
Matplotlib • Seaborn

Introduction :

Breast cancer is one of the most common and life-threatening diseases among women worldwide and when it is discovered and diagnosed in the early stages it'll help improve treatment outcomes and survival rates.

In this project we will be applying Machine Learning models to predict whether a tumor is benign or malignant based on cell feature measurements and to be able to identify the most accurate model that can support medical decision-making.

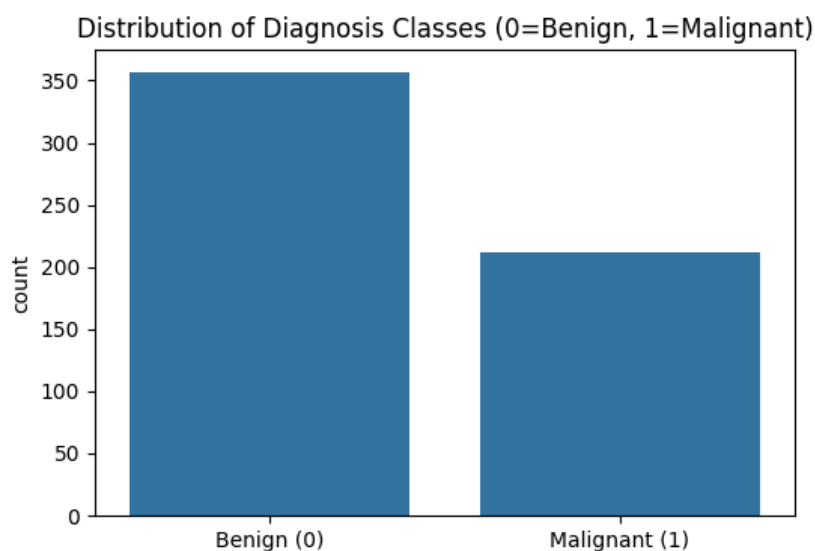
Data Overview:

- This dataset consisted of 560 records and over 30 numerical features representing cellular characteristics of the tumor cell properties such as radius_mean (Average radius of tumor) , texture_mean (variation in texture) ,area_mean (Size of tumor) and etc..... These measurements help the model distinguish between benign and malignant tumors.
- Source: Kaggle (CSV file)
- Target variable: diagnosis (and was transformed into binary classification by Label encoding where the 0 represent 'Benign' and the 1 represent Malignant).
- After executing the pandas there's no missing values in any column and the Unnecessary columns such as *id* and *Unnamed: 32* were removed .
- We also added feature engineering to improve representation: total_mean,compact_ratio,log_area

And tested performance on another dataset (Iris) to study model behavior across domains.

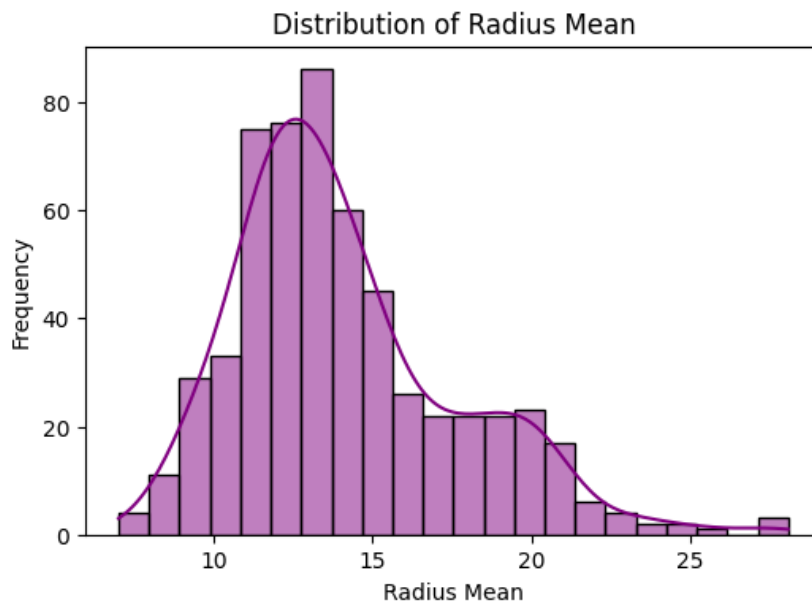
EDA:

1- Class Balance



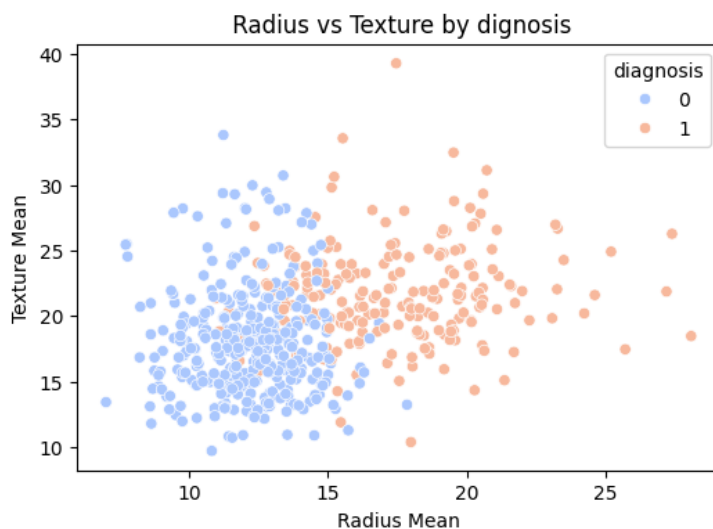
This chart describes the numbers 0 for benign and 1 for malignant tumors in the dataset , and it's clear that the benign is slightly more common than the malignant but distribution is still acceptable for training machine learning models without major imbalance concerns.

2- Feature Distributions



This histogram shows how the Radius mean feature is distributed across the chart, it appears a little bit skewed and that means that some tumors may have larger radius values which can help differentiate between benign and malignant cases.

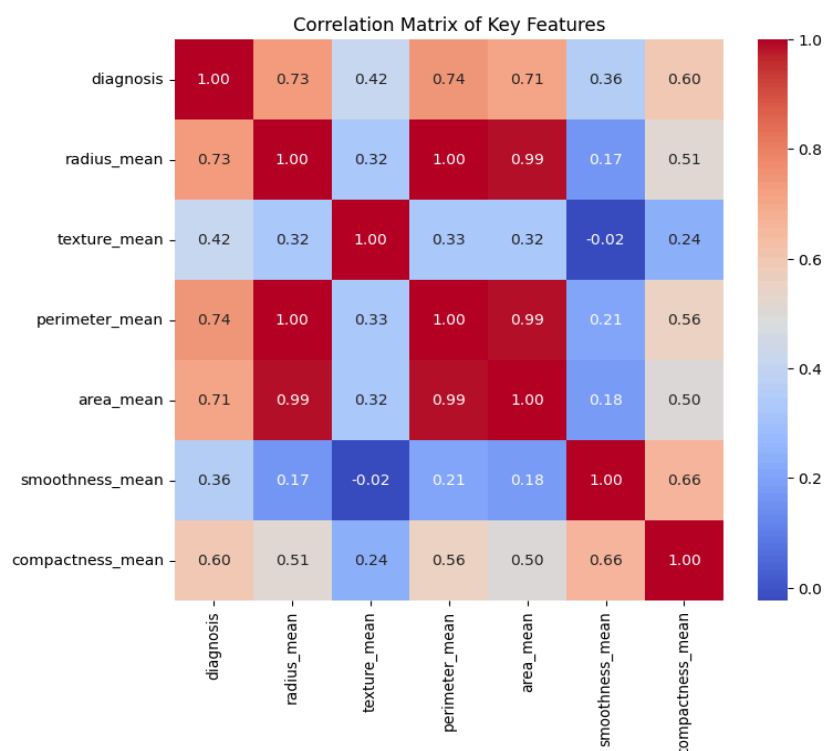
3- Feature Relationships



This Scatter plot indicates how Radius and Texture differ between benign and malignant cases where the malignant case appear to be in higher values than the benign case ,that's means that the Radius and the Texture features helps to distinguish between the two cases

4- Correlation

Correlation analysis helps measure how strongly features are related to each other and to the target variable.



This heatmap shows the relationships between tumour features and the diagnosis . For example, how the radius_mean ,the perimeter_mean and the area_mean have strong correlation with the malignant tumors which means larger tumor size is strongly linked to cancer severity, also smoothness and compactness have a moderate correlation which helps improve prediction accuracy.

Methodology:

Several machine learning algorithms were used to build a classification model that can predict whether the tumor is benign or malignant. The overall methodology followed these steps:

1-Data Preparation

- Removed unnecessary columns.

- Verified that no missing values existed in the dataset.

- converted the variable *diagnosis* into numerical values where 0 represents benign and 1 represents malignant

- Split the dataset into:

 - Training set: 70%

 - Testing set: 30%

2-Exploratory Data Analysis (EDA)

- Feature Distribution

A histogram was plotted for the *radius_mean* feature to observe its distribution and identify any skewness, outliers, or irregularities. Kernel Density Estimation (KDE) was added to visualize the underlying probability distribution.

- Feature Relationships

A scatter plot between *radius_mean* and *texture_mean* was created, with points colored based on the diagnosis (Benign = 0, Malignant = 1). This helped reveal how tumor size and texture vary across the two classes and whether they exhibit separation patterns.

- Class Imbalance Check

A count plot of the diagnosis variable was generated to examine class distribution.

- Correlation Analysis

A correlation heatmap was produced focusing on key features, including *radius_mean*, *texture_mean*, *perimeter_mean*, *area_mean*, and others. This helped identify highly correlated features and understand which characteristics are most related to the malignancy of tumors.

3-Feature Scaling

Some models do not work well when the numbers in the features are very different from each other. Because of that **StandardScaler** was applied to normalize the numerical features for these models.

This step is important for **Logistics regression** and **SVM** because these models are sensitive to value differences. While tree models like **Random forest** and **Gradient boosting** do not need scaling, since they can handle different value sizes without any problem.

4-Machine Learning Models

1-Logistic Regression

Simple linear model used for binary classification tasks. It works by finding a straight line that separates the two classes.

2-Support Vector Machine (SVM – RBF kernel)

SVM is good at handling more complex data where the classes are not separated in a straight line. The RBF kernel helps the model draw curved boundaries, which makes it useful for medical datasets.

3-Random Forest Classifier

Made of many decision trees that work together, it tends to be more stable and less likely to overfit.

4-Gradient Boosting Classifier

Builds decision trees one at a time, and each new tree tries to fix the mistakes of the previous one. This makes the model strong and able to capture patterns in the data accurately.

5- Artificial Neural Network (ANN)

ANN is a deep learning model inspired by how the human brain works. It learns patterns by passing data through multiple connected layers of neurons.

In this project, the ANN contains an input layer, one hidden layer, and an output layer.

It is useful for capturing complex, non-linear relationships in the dataset, making it powerful for medical diagnosis tasks like detecting malignant tumors.

5-Model Evaluation

Each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC–AUC Score

The goal was to identify the most accurate and reliable model to diagnosing tumors.

6-Multi-Domain Dataset Comparison

To evaluate the generalization ability of the models across different domains, both Logistic Regression and Random Forest were trained and tested on two datasets: the Breast Cancer Wisconsin Diagnostic dataset and the Iris dataset (converted to binary classification). The results show consistent high performance across both datasets.

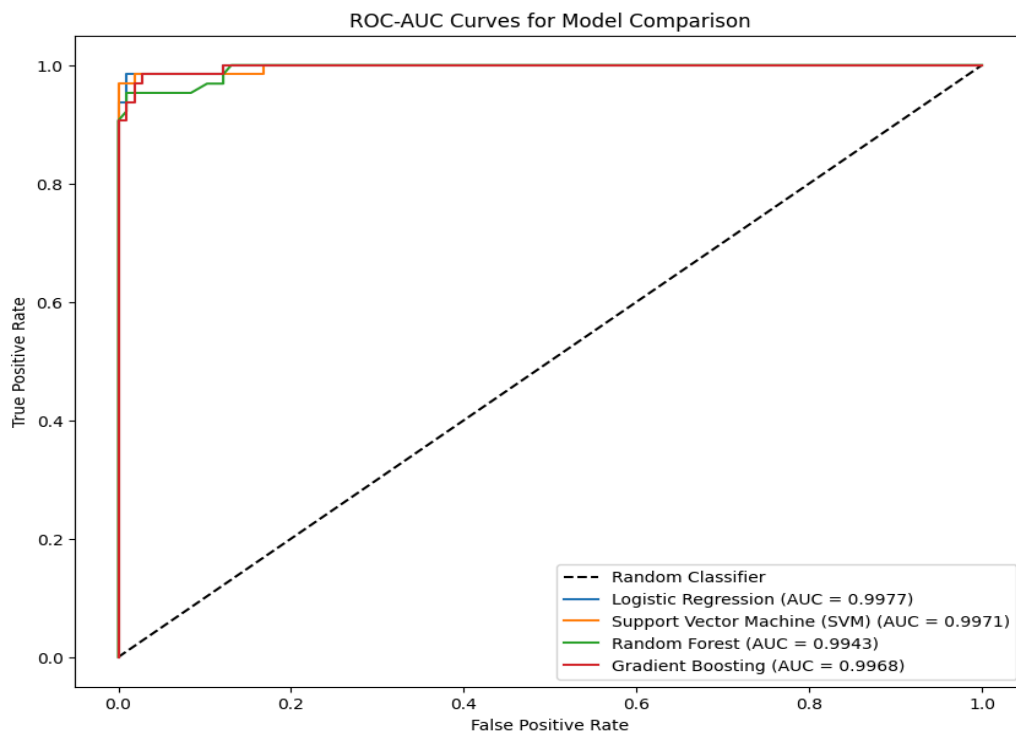
For the Breast Cancer dataset, Logistic Regression achieved an accuracy of 0.9766, slightly outperforming the Random Forest model, which achieved 0.9649. This indicates that Logistic Regression was more effective at separating malignant and benign cases in this dataset, likely due to the linearly separable nature of many breast cancer features.

On the Iris dataset, both models performed exceptionally well. Logistic Regression achieved 100% accuracy, while Random Forest also reached 100% accuracy. These results reflect the simplicity and strong class separation of the binary Iris classification used in this evaluation.

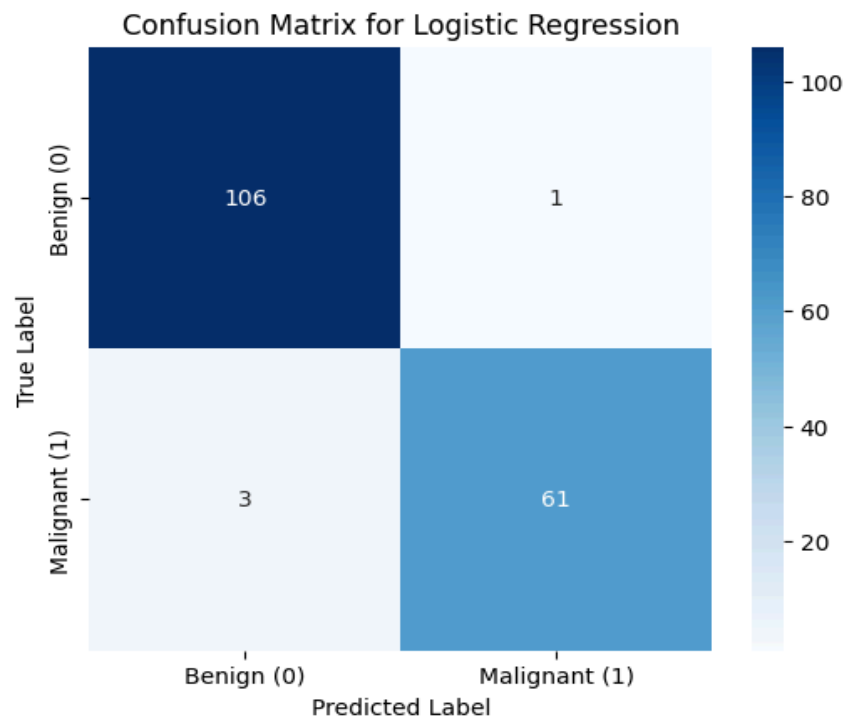
Results:

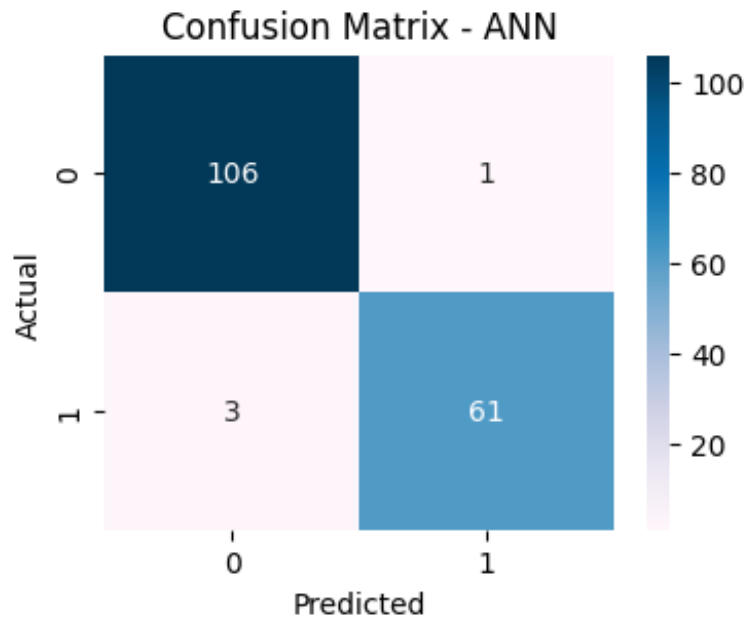
Best Performing Model:

After training and testing all four models ,both **SVM** and **Random Forest** achieved the highest metrics, each reaching around 97.4% accuracy with excellent AUC scores close to 0.99.



confusion matrix:





From the confusion matrix for both LR and ANN, the models were able to correctly classify most benign and malignant cases. Only a few predictions were incorrect, indicating that the models can clearly separate the two classes. This aligns with the high ROC–AUC score, demonstrating that the models are reliable in distinguishing between benign and malignant tumors.

How to Excel:

- **1- Apply feature engineering to improve performance:**

We created additional features from existing ones to provide the models with more informative patterns

Example features engineered:

- **total_mean** = combination of radius, texture, perimeter, and area
- **compact_ratio** = compactness_mean / radius_mean
- **log_area** = logarithmic transformation of area to reduce skewness

These engineered features helped the models better differentiate between benign and malignant tumors by enriching the dataset with more useful insights.

2- Hyperparameter Tuning in Deep Learning

This section describes the parameters and hyperparameters used for each machine learning model implemented in the Breast Cancer Wisconsin (Diagnostic) dataset analysis, as well as the neural network architecture. All models were trained using a controlled experimental setup with a fixed random state (42) to ensure reproducibility.

Data Preprocessing Parameters

Label Encoding

Technique: LabelEncoder

Mapping:

Benign (B) $\rightarrow$ 0

Malignant (M) $\rightarrow$ 1

Feature Scaling

- Scaler: StandardScaler
- Used for: Logistic Regression, SVM, ANN
- Rationale: Models relying on distance or gradient optimization require standardized feature scales.

Train/Test Split

Training: 70%

Testing: 30%

Stratification: Yes (stratify=y)

Random State: 42

Logistic Regression

No manual hyperparameter tuning was applied. The model used default regularization parameters, appropriate for binary classification on medium-sized datasets.

Support Vector Machine (SVM)

Default SVM settings with RBF kernel provide strong nonlinear classification without explicit tuning.

Random Forest Classifier

The model uses typical baseline settings commonly used for structured tabular datasets.

Gradient Boosting Classifier

The model uses standard hyperparameters for balanced datasets.

Artificial Neural Network (ANN)

ANN Architecture

Layer Type	Number of Units	Activation function	Notes
Input Dense Layer	32	ReLU	input shape = number of features
Hidden Dense Layer	16	ReLU	Fully connected
Output Layer	1	Sigmoid	binary classification output

ANN Training Hyperparameters

Hyperparameter	Value	Description
Optimizer	Adam	Adaptive optimization algorithm
Loss Function	Binary Crossentropy	For binary classification
Metric	Accuracy	performance measurement
Epochs	100	Maximum training cycles
Batch Size	16	Samples per update
Validation Split	0.2	20% of data for validation

Early Stopping Configuration

Parameter	Value	Purpose
Monitor	val_loss	Track validation loss
Patience	10	Stops if no improvement after 10 epochs
Restore Best Weights	True	Use best model weights

3-Try Multiple Dataset Domains for Performance Comparison:

Dataset :Breast Cancer Domain: Medical

Dataset: Iris (binary) Domain: General

This allowed us to observe:

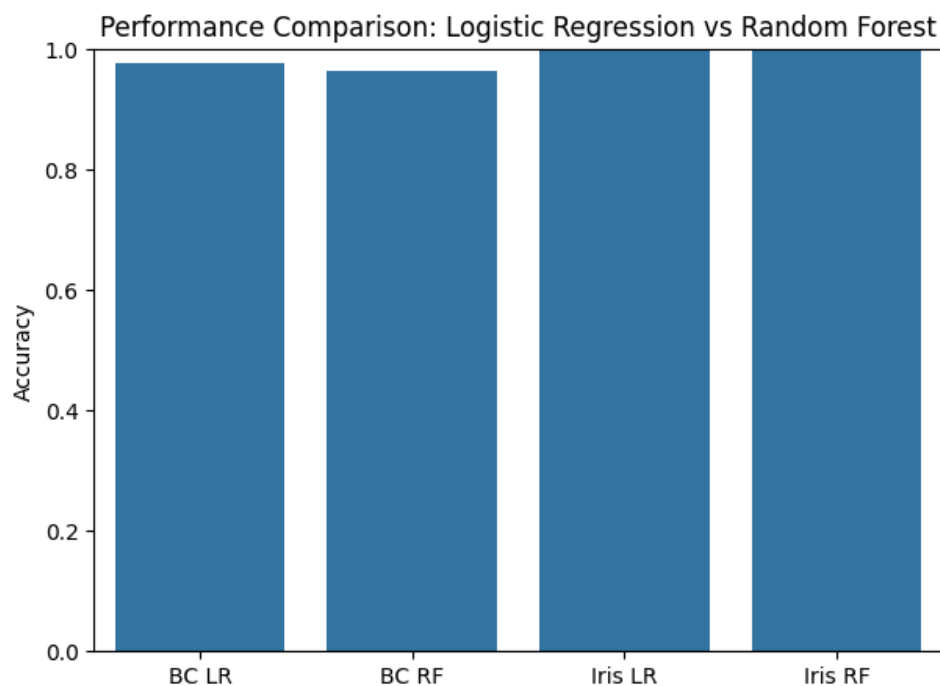
- Whether the same models perform consistently across different problem types:

Both Logistic Regression and Random Forest achieved high accuracy on both datasets, showing consistent performance across domains.

- How dataset characteristics affect classification accuracy:

Logistic Regression slightly outperformed Random Forest on the Breast Cancer dataset (0.9766 vs 0.9649), likely due to the linear separability of features. On the simpler binary Iris dataset, both models achieved 100% accuracy, highlighting that simpler or well-separated data leads to higher classification performance.

The bar-chart visualization further highlights the close performance between the two models across both datasets, with all accuracies falling between 0.96 and 1.00. Overall, the comparison demonstrates that both models are robust, with Logistic Regression showing a slight edge on the breast cancer prediction task.



Conclusion:

This project explored how different machine learning models perform on a tumor classification problem using a real medical dataset. After completing all preprocessing steps, building the models, and evaluating them using accuracy, precision, recall, F1-score, and ROC-AUC, the results showed that the best-performing model was able to separate benign and malignant tumors clearly with very few errors.

Overall, this project followed the full machine learning workflow from data exploration to modeling and evaluation, showing that even simple ML methods can achieve strong results when applied correctly.